KEOLEBOGILE GILLIAN Segwagwa
Brightlight Data Analytics
Assignment No: Research Assignment 01
Due date: 16 April 2026

Section A: Data Foundations

1. What is a Database?

A database is an organized collection of data that is stored electronically and can be easily accessed, managed, and updated.

Databases help store information in a structured way so that users and systems can quickly retrieve and analyze data.

Why databases are important
Database are essential in the modern world because:
- They improve decision making
- They allow fast access to data
- They store large amounts of information

Real world examples
- Schools store students records, attendance and marks.
- Banking systems store customer account information and transactions.

2. What is a Data Warehouse?

A Data Warehouse is a large storage system designed to collect and store data from multiple sources for reporting and analysis.

Difference Between Database and Datawarehouse

Database - Stores current operational data
Datawarehouse - Stores historical data

Database - Used for daily operations
Data Warehouse - Used for reporting and analytics

Database e.g. Processing orders
Data warehouse e.g. Sales trends over 5 years

Why companies use Data Warehouses

- They use it to
  * Identify trends
  * Analyze business performance
  * Make strategic decisions

3. What is a Data Lake?

A data lake is a storage system that holds large amounts of raw data in it's original format until it is needed.

Data Lake Vs Data Warehouse

Data Lake	Data Warehouse
* Stores raw data	Stores processed data
* Supports structured and unstructured data	Mostly structured data
* Used by data scientists and analysts	Used for business reporting

Business use data lakes when:

* They collect large amounts of data
* They want flexibility
* They use AI and Machine Learning

4. What is a Data Mart?

A data Mart is a smaller section of a data warehouse focused on a specific department or business area.

example: Finance Data Mart, Sales Data Mart, Marketing Data mart.

Who uses Data Marts: Managers, Business Analysts Department teams.

7. What is Data Mining
is the process of analyzing large datasets to discover patterns and insights

Real world e.g : Retail * Stores analyse purchase patterns
Banking * Detect fraud transactions

8. What is Data base Management System (DBMS)

is a software used to create, manage and interact with databases.

Popular DBMS - MySQL * Used for websites
* Open Source
- PostgreSQL * Used for analytics
* Advanced open source
- Oracle * Enterprise database
* Used by large corporations
- MongoDB * NoSQL database
* Used for flexible data storage

9. What is ETL?
ETL stands for:
<u>Extract</u> - Collect data from sources
<u>Transform</u> - Clean and format data
<u>Load</u> - Store into database or warehouse

Why ETL is important
- Enables analytics
- Improves data quality
- Combines multiple data sources

5. Structured Vs Semi-Structured Vs Unstructured Data.

Structured data
Highly organised data stored in rows and columns
e.g SQL tables and excel spreedsheets

Semi-structured data
Data that does not follow strict tables but still has structure e.g XML files and JSON files.

UnStructured data
Data without a defined structure e.g. Videos, Images, emails.

6. What is Data Modeling?

Data modeling is the process of desiguing how data will be stored, organised and related in a system.

Why it is important?
- Reduces errors
- Makes systems easier to maintain
- Improves performance.

Types of Data Models : Conceptual model * High-level overview
* Shows relationships

: Logical Model * More detailed
* Defines tables & relationships

: Physical Model * Actual deploy

10. Data Storage Formats

CSV * Simple text File
* Easy to use

JSON * structured text Format
* Used in APIs

Parequet * Column-based Format
* Efficient for analytics

Avro * Compact binary Format
* Used in big data

Section B: Artificial Intelligence & Machine learning

11. What is Artificial Intelligence?

AI is the ability of machines to perform tasks that normally require human intelligence.

Narrow AI * Chatbots
* It is designed for one task

General AI * Human-level intelligence e.g Not yet fully developed.

12. What is Machine learning?

Machine Learning is a type of AI where systems learn from data instead of being programmed.

Types of Machine learning
* Supervised Learning - use labeled data
* Unsupervised Learning - Finds patterns
* Reinforcement Learning - Learns from rewards

13. What is Deep Learning?
Is a subset of machine learning using neural networks
Neural networks * Layers of algorithms
* Learn patterns.

14. What is NLP?
Natural Language processing allows computers to understand

human language
e.g ChatGPT, Voice assistants, Email Spam Filters

15. What is Large Language Models (LLM)
are AI systems trained on large text datasets
e.g. ChatGPT, Claude

16. What is Generative AI?
it creates content like text, images or code.
e.g DALL-E, Midjourney, ChatGPT

17. What is Training Dataset?

is data used to train AI models
Poor Data Causes * Bias
* Wrong predictions

18. What is Computer Vision?
It allows machines to interpret images.
e.g Face recognition and self-driving cars.

Section C; APIs & Developer Concepts

19. What is API?
An API allows systems to communicate.
e.g waiter in restaurant. it enables seamless integration
enhance Functionality and improve user experiences

20. What is API Key?
API Key is a unique identifier
Purpose - Security

21. What is Rest API?
REST stands for Representational State Transfer
GET * Retrieve data
POST * Send data
PUT * Update data
Delete * Remove data.

22. What is JSON?
JSON is data format e.g {"name": "John", "age": 21}

23. Prompt Engineering
Poor Prompt ; Explain AI
Better Prompt: Explain AI in Simple terms For beginners
with examples.

24. MAP Function

Python e.g: numbers = [1, 2, 3] result = List(Map
lambda x: x * 2, numbers))

25. Cloud computing
IaaS * AWS EC2
Paas * Google App Engine
Saas * Gmail

26. Version control
Tracks changes to a file over time
GIT tracks changes
Commit - Save changes
Branch - New version

27. Jupyter Notebook
Used with Python
Popular because * Interactive and easy visualization

28. Python
Popular for analytics It's libraries are * Pandas
* Numpy
* Matplotlib
* Scikit-learn

Pendas

32. What is Bias in AI

Refers to the occurence when artificial intelligence systems produce unfair or skewed outcomes due to human biases embedded in training data, model design, or deployment processes.

AI Models learn from historical data. If the data contains bias, the AI will learn and repeat that bias. This can lead to unfair decisions in areas like hiring, banking healthcare and law enforcement.

Real world examples of AI Bias is Amazon's AI hiring tool.

Potential Consequences of AI Bias

* Unfair hiring decisions
* Discrimination against certain groups.
* Loss of trust in AI systems
* Legal and ethical problems for companies.
* Reduced diversity in workplaces

AI Bias is a serious issue because AI is increasingly used to make important decisions affecting people's lives.

Section D: Data Ethics

29. Data Privacy
Data privacy protects personal information.
GDPR – European regulation

30. Data Ethics?
It tells an organization whether it has been using it's data ethically and correctly. It guides organizations regarding the usage of data, machine learning algorithms in AI and other technologies to avoid bias.

31. Data governance?

is the discipline that ensures the quality, security and availability of an organization's data by defining policies, standards and procedures for data management.

31. 1. Data governance Framework
Outlines how data governance will be implemented including policies
31. 2. Roles & Responsibilities
defines roles such as Data Owners, Data stewards.
3.1 3. Policies & Procedures
Dictate how data should be handled including data classificat

33. Differences Between Data Roles

Role	Main Responsibility	Key Skills	Common tools
Data Analyst	Analyzes data and Creates reports	Excel, SQL Visualization	Excel, Power BI, Tableau
Data Scientists	Builds Predictive models and Finds Patterns	Statistics, Python Machine learning	Python, R, Jupyter scikit-learn
Data Engineer	Builds data pipelines and manages data infrastructure	Programming Databases Cloud	SQL, Python Spark, Databricks.
AI Engineer	Builds and deploys AI models and applications	Machine learning, Deep Learning	TensorFlow, PyTorch, Python

Data Analyst - looks at data and explains what happened

Data Scientist - Predicts what will happen

Data Engineer - Builds systems to move and store data

AI Engineer - Builds intelligent AI systems

34. What is Business intelligence (BI)?

refers to technologies and processes used to collect analyze and visualize business data to help organizations make better decisions. BI combines data analytics, reporting, dashboards and visualization tools to turn data into insights

Popular BI Tools

Power BI tools - Created by Microsoft
- Creates dashboards and reports

- helps track business performance
- Used by managers and executives

2. Tableau
- Data visualization tool
- Creates interactive charts and dashboards
- Helps identify trends and patterns

Business uses BI tools to:

- Monitor sales performance
- Track customer behavior
- Improve operations
- Make data driven decisions

SECTION E: Personal Reflection

35. Which concept did you find most interesting

The AI and AI Bias. It was surprising to learn that AI systems can make unfair decisions based on the data they are trained on.

36. Which concept did you find most difficult

Understanding the difference between Data Engineer Data scientist and AI Engineer. At first, they all seemed similar because they all work with data and AI.

37. How do you see data and AI affecting your Future?

References:

1. Tableau. (2024). What is Business Intelligence? Tableau. https://www.tableau.com/Learn/articles/Business-Intelligence.

2. TechTarget. (2025). AI Engineer vs Data Scientists. TechTarget. http://www.techtarget.com

3. Fertev, X. (2020). Bias and Discrimination in AI. arXiv.org. https://arxiv.org

4. Langenkamp, M. (2020). Hiring fairly in the Age of Algorithms. arXiv.org. https://arxiv.org

5. Microsoft. (2024). Power BI Overview. Microsoft learn. https://learn.microsoft.com

6. IBM. (2024). What is Artificial Intelligence Bias? IBM. https://www.ibm.com

7. Google Cloud. (2024). Data Analytics Roles Explained. Google Cloud. https://cloud.google.com

8. BrightLearn. (2026). Data Analytics Course Material. BrightLearn

Data and AI will play a major role in my future career and everyday life.

- I plan to move into Data Analytics
- I will help businesses make better decisions
- AI tools will help automate analysis and improve productivity

R